MODULE 10

MSK Soft Tissue & Bone Tumors: A Systematic Imaging Approach

Lipoma vs. Liposarcoma · PVNS · USP6 Family · BNCT · Osteosarcoma · Sarcoma Red Flags

1.5 AMA PRA Cat 1 | **Soft Tissue Tumors** | **CME Article Review**

LEARNING OBJECTIVES

1. Apply WHO 2020 classification principles to common MSK bone and soft tissue tumors.
2. Distinguish lipoma from well-differentiated liposarcoma using MRI criteria.
3. Recognize the USP6 family of benign neoplasms (myositis ossificans, ABC, nodular fasciitis).
4. Identify PVNS/TSGCT using low T1/T2 signal and GRE blooming artifact.
5. Apply BNCT diagnostic criteria and understand why biopsy is NOT recommended.
6. Recognize imaging red flags for malignancy: size, enhancement, edema, bone destruction.

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1. Systematic Approach — Malignancy Red Flags

Feature	Concerning	Benign Favoring
Size	>5 cm (soft tissue); >3 cm (bone)	<3 cm; stable on serial imaging
Enhancement	Avid, solid, nodular	None (lipoma, cyst, BNCT) or thin rim only
Signal	Heterogeneous; necrosis; hemorrhage	Homogeneous; simple cyst-like
Margins	Infiltrative; ill-defined; fascial transgression	Well-defined; encapsulated; pseudocapsule
Bone	Cortical disruption; endosteal scalloping >2/3 width	Intact cortex; sharp zone of transition
Perilesional edema	Extensive T2 STIR edema around lesion	Absent or minimal
Periosteal reaction	Codman triangle; spiculated; sunburst	Solid, continuous (reactive bone)

CRITICAL

A >5 cm deep soft tissue mass with solid enhancement and perilesional edema = sarcoma until proven otherwise. Refer to a dedicated bone and soft tissue tumor center BEFORE biopsy to avoid seeding or compromising surgical margins.

2. Lipomatous Tumors — Lipoma vs. Liposarcoma

MRI Discrimination: Lipoma vs. WDLPS vs. DDLPS

Feature	Lipoma	WDLPS	DDLPS (Dedifferentiated)
Fat content	Homogeneous; >90% fat	Predominantly fat; admixed	Non-adipose nodule(s) within fat
Septa	Thin, regular (<2 mm)	Thick, irregular (>2 mm)	Thick septa + solid non-fatty component
Internal nodules	Absent	May have small foci	Large enhancing non-fatty nodule
Enhancement	None	Minimal/thin septa only	Solid nodular enhancement in dediff component
Key discriminator	Pure fat; thin septa; no enhancement	>2 mm septa or non-fat component	Enhancing non-fat mass within lipomatous tumor

CRITICAL

Deep lipomatous mass with ANY thick (>2 mm) irregular septa, non-fatty nodule, or enhancement → cannot call it a lipoma. Report: "Lipomatous neoplasm; WDLPS cannot be excluded; recommend MDT review and excision."

3. USP6 Family, PVNS/TSGCT & BNCT

USP6 Family — Benign Entities That Mimic Malignancy

Entity	Location	Imaging Hallmark	Clinical Clue
Myositis ossificans	Intramuscular (thigh, arm)	Peripheral ossification (zonal phenomenon); no rim enhancement	History of trauma; self-limiting; rapid growth
Aneurysmal bone cyst (ABC)	Metaphysis; any bone	Fluid-fluid levels on T2 MRI; expansile	Adolescent; rapid expansion
Nodular fasciitis	Superficial; fascial	Small (<3 cm); T2 bright; rapid growth (weeks)	Wrist/forearm most common
Fibroma of tendon sheath	Adjacent to tendon (wrist/hand)	Low T1/T2; lobulated; attached to sheath	Slow growing; finger/wrist

PVNS / TSGCT — MRI Characteristics

Feature	Finding	Explanation
T1 signal	Low to intermediate	Hemosiderin; fibrous tissue; lipid-laden macrophages
T2 signal	LOW (characteristic)	Hemosiderin = paramagnetic; suppresses T2
GRE sequence	BLOOMING artifact (most sensitive)	Hemosiderin amplifies local field inhomogeneity
Enhancement	Heterogeneous; synovial thickening	Avid villonodular synovitis
Distribution	Diffuse (PVNS) vs. Focal (TSGCT nodule)	Diffuse = joint replacement risk; focal = arthroscopic excision

BNCT vs. Chordoma — MRI Discrimination

Feature	BNCT (Benign)	Chordoma (Malignant)
T1 signal	Hypointense (↓)	Hypointense (↓)
T2 signal	Hyperintense (↑)	Hyperintense (↑); "bubbly"
Enhancement	NONE (key discriminator)	PRESENT — avid, lobular
Bone destruction	ABSENT (intraosseous)	Present; cortical disruption
Soft tissue mass	ABSENT	Present in most cases
Management	Surveillance; NO biopsy	Surgery ± proton beam RT

CRITICAL

When BNCT features are typical: state in your report "Imaging features are most consistent with benign notochordal cell tumor. Biopsy is NOT recommended for this typical appearance; imaging surveillance is appropriate." Atypical features (lysis, soft tissue extension, enhancement) → ANCT → further evaluation warranted.

4. Periosteal Reaction Patterns

Pattern	Appearance	Typical Diagnosis
Solid/continuous	Dense uniform new bone; no interruption	Reactive (stress, infection, healing)
Laminated ("onion skin")	Parallel concentric rings	Ewing sarcoma; osteomyelitis
Spiculated ("sunburst")	Radiating from cortex outward	Osteosarcoma; Ewing (secondary)
Codman triangle	Periosteal elevation at margins; triangular lucency	Osteosarcoma; Ewing; aggressive infection
Zonal (peripheral)	Mature periphery; soft center; outside-in	Myositis ossificans — BENIGN; USP6 family

References

- Kransdorf MJ et al. Radiology 2002.
- Broski SM, Wenger DE. Skeletal Radiol 2023.
- Yamaguchi T et al. Skeletal Radiol 2008.